MSG No. 107.232 I_c4mm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,5,6,26,27,31,32]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,7,8,25,28,29,30]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,12,13,14,18,19,23,24]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,15,16,17,20,21,22]`

Table 3: Wyckoff site: `8c`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,8,28,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[2,5,27,30]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[3,6,26,29]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[4,7,25,32]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[9,16,20,23]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[10,13,19,22]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[11,14,18,21]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[12,15,17,24]`

Table 4: Wyckoff site: `16d`, site symmetry: `.m.`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,5]`
2	$[-y,\ 0,\ z]$	`[2,7]`
3	$[y,\ 0,\ z]$	`[3,8]`
4	$[0,\ -y,\ z]$	`[4,6]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,13]`
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,15]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,14]`
9	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[17,21]`
10	$\left[-y,\ 0,\ z+\frac{1}{2}\right]$	`[18,23]`
11	$\left[y,\ 0,\ z+\frac{1}{2}\right]$	`[19,24]`
12	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[20,22]`
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[25,29]`

continued ...

Table 4

No.	position	mapping
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z\right]$	[26,31]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z\right]$	[27,32]
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[28,30]

Table 5: Wyckoff site: 16e, site symmetry: ..m

No.	position	mapping
1	$[x,\ x,\ z]$	[1,8]
2	$[-x,\ x,\ z]$	[2,5]
3	$[x,\ -x,\ z]$	[3,6]
4	$[-x,\ -x,\ z]$	[4,7]
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9,16]
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10,13]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[12,15]
9	$\left[x,\ x,\ z+\frac{1}{2}\right]$	[17,24]
10	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[18,21]
11	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	[19,22]
12	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	[20,23]
13	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[25,32]
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[26,29]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[27,30]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[28,31]

Table 6: Wyckoff site: 16f, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,32]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,29]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,30]
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[4,31]
5	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[5,26]
6	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[6,27]
7	$\left[\frac{1}{2}-x,\ -x,\ z\right]$	[7,28]
8	$\left[x+\frac{1}{2},\ x,\ z\right]$	[8,25]
9	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[9,24]
10	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10,21]
11	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11,22]
12	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[12,23]
13	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	[13,18]
14	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[14,19]
15	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15,20]
16	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16,17]

Table 7: Wyckoff site: 32g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[-x,\ y,\ z]$	[5]
6	$[x,\ -y,\ z]$	[6]
7	$[-y,\ -x,\ z]$	[7]
8	$[y,\ x,\ z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[21]
22	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[23]
24	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[26]
27	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[29]
30	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[31]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[32]